Literal structural certificates for the $U_3(5)$ family

Private L054 finite application

Theorem 1. *Relative to the acquired numerical and named-model inputs, the displayed structural assertions hold for all 565 rows of the four current companion tables: 80 for $U_3(5)$, 93 for $U_3(5) : 3$, 148 for $U_3(5) : 2$, and 244 for $U_3(5) : S_3$. Each expression asserts its indicated decomposition, without claiming that the expression determines a unique isomorphism type.*

Source and actual inputs. The current Connor–Leemans PDFs are dated November 26, 2012. All 24 native pages were directly inspected. Baseline $24 = C_{24}$ in rows 3 and 50 of `pgu35`, and rows 7 and 144 of `pgammau35`, is distinguished from the superscript 2^4. The powers $6^2 = C_6^2$, $5^2 = C_5^2$, and 5^{1+2} have their separate meanings. Explicit right parentheses in $5^{1+2} : (8 : 2)$ and $5^{1+2} : (2 \times 4)$ are preserved. Default left association applies to $5^{1+2} : 4 \cdot S_3$ and $2 \cdot A_5 \times 3 : 2$. Colons and nonsplit dots are not exchanged.

The accepted robust weights and normal/arbitrary subgroup restrictions supply the complete actual subgroup inventories and source-row bijections. All four original actions have degree 651. Their retained group files or complete states are parsed as exact permutation generators; displayed constructor names never reconstruct a proper embedded subgroup. The corrected `pgammau35` weights and its distinct review are acquired, including the retained earlier duplicate-local-class failure. No numerical census or source-pair comparison is repeated here.

The original clean `PSU(3,5)` execution and all its recorded pins are acquired. Its ordered generators equal the retained top generators. GAP Reference Manual 50.2-14 is the explicit named-constructor premise. A literal invariant 126-point subset is checked for each action, including every actual subgroup generator. The four image orders are respectively 126000, 378000, 252000, and 756000. The acquired invariant-subset lemma therefore makes each restriction an isomorphism, not merely an order match. This supplies the new named $U_3(5)$ model; the three larger named-model interfaces remain those of the accepted numerical constructions.

Lemma 2 (Exact compact encoding and generated maps)**.** *A permutation on $d \leq 256$ letters can be stored by its d image bytes. For right actions, multiplication is $ab = (b_{a_i})_i$. Let $M = \langle g_1, \ldots, g_k \rangle$ be an acquired finite model and let h_j be permutations of another set. Enumerate M by its generator edges from the identity. Assign the identity image, propagate images by right multiplication by h_j, and check every repeated edge assignment. If the resulting map is bijective onto an actual group H, it is an isomorphism $M \longrightarrow H$ taking g_j to h_j.*

Proof. Byte encoding is injective on the finite image arrays. Replacing each byte a_i by table entry b_{a_i} gives exactly the displayed multiplication; unused table entries do not affect values below d. The implementation uses this representation only for $65 \leq d \leq 256$ and ordinary tuples otherwise. Decoding preserves equality, inversion and all group operations. For the map, every element is reached because the model generators generate. Every edge is checked, including edges not used for first discovery. Induction on a generator word gives $f(xw) = f(x)w(h_1, \ldots, h_k)$. Taking $x = 1$ and then arbitrary x proves $f(xy) = f(x)f(y)$. The checked bijection proves the assertion. Thus storing the ordered generator images instead of every map value removes no edge or bijection condition from the acquired MC.3 test. □

Two explicit small atoms. For C_6^2, the reader checks two commuting elements of order 6, generation, and group order 36. The resulting surjection $C_6 \times C_6 \to H$ has equal finite orders and is an isomorphism. For 5^{1+2} it checks $|H| = 125$, $H = \langle x, y \rangle$, a central commutator $z = [x, y]$ of order 5, $Z(H) = \langle z \rangle$, and $x^5, y^5 \in \langle z \rangle$. Hence $H' = \langle z \rangle$ and $H/Z(H) \cong C_5^2$. Every index-five maximal subgroup contains H', while the inverse images of all lines of $H/Z(H)$ intersect in $Z(H)$. Therefore $\Phi(H) = H' = Z(H)$ has order 5, which is precisely the asserted extraspecial structure. The criterion prescribes no exponent and infers no unprinted extraspecial sign.

The $2 \cdot A_5$ factor. The acquired central-commutator criterion applies unchanged. The reader checks the actual central kernel $N = C_2$, all 60 quotient cosets, their literal action, and its generator-edge isomorphism with the acquired A_5 model. It checks $[x, y] \in N \setminus \{1\}$. The quotient images commute, but all four pairs of lifts in $Nx \times Ny$ have this same nonidentity commutator. A complement would supply commuting lifts, a contradiction. This criterion was previously proved with root's contribution; its present application claims no independent authorship of that argument.

All other structural maps. The registry reuses the previously accepted explicit models, including A_7, A_6, M_{10}, M_9, and $L_2(7)$. Every new extension model has an actual normal factor, and the reader checks its transported generators and normality inside the actual model. A split extension has an explicit complement with trivial intersection and the product of orders equal to $|H|$. The displayed nontrivial action has a noncommuting generator pair. A direct product additionally has commuting factors. Prime-index nonsplitting checks every outside lift. The remaining fixed small quotients use the acquired complete coset-normal-form and generator-lift criteria. Every source occurrence then receives a separately checked isomorphism to its exact retained embedded subgroup, under the same 126-point restriction map.

Finite evidence and preserved failures. The first bounded witness pass stopped at an unsupported A_5 quotient. The second completed but left four explicit missing certificates because of the erroneous baseline-24 token interpretation. No failed row was accepted. Exact prior inputs and receipts remain, with hash-bound historical aliases. The third pass, after the source-specific correction, completed all 565 candidates in 14.45 seconds. GAP proposes witnesses only; the full ordinary and optimized Python checks prove their literal properties and must return identical reports. Neither mode repeats a subgroup census.

Independent controls compare 3456 products of all S_4 permutations at six encoding-boundary degrees, and compare a complete generated S_3 map with an independently constructed explicit map. They reject incompatible generator images. Explicit C_6^2 and Heisenberg order-125 models test the two new atoms, with wrong-order, missing-generation and commutator damages. The actual orbit validator rejects a nonfaithful invariant subset, while a faithful diagonal example passes. Source punctuation, baseline 24, right parentheses, embeddings, complete states, and the named top generators have focused damages. All earlier extension and map controls are acquired.

The frozen report gives the exact registry, map, edge, lift and guard counts. This is a current-table certificate, not proof of historical PDF byte identity or a canonical naming algorithm. It grants no whole-original, question, formalization or measurement credit. Authorship of the application, compact implementation and its elementary encoding proof is `1044_resume`; prior mathematical authorship and distinct reviews remain explicit. Root conducts the new distinct review.